

Binomial random variables



1 In a binomial experiment:

- a Are the trials independent or dependent?
- b How many possible outcomes are there for each trial?

2 State whether following scenarios describe a binomial random variable:

- a Rolling a six-sided die 20 times and counting the number of times the die lands on 1.
- b The sum of the outcomes of rolling a six-sided die 10 times.
- c Selecting 70 random people from the population and recording the number of females.
- d A particular coin is more likely to land on heads than tails. Tossing this coin 10 times and recording the number of times it lands on heads.
- e Drawing a marble with replacement from a bag containing purple, green and yellow marbles, and noting the number of purple marbles.
- f Drawing a marble with replacement from a bag containing purple, green and yellow marbles, and noting the number of red marbles.

3 Consider the binomial probability formula:

$$P(x) = {}^nC_x p^x (1-p)^{n-x}$$

where $P(x)$ is the probability of obtaining x successes in n independent trials.

- a What does p represent in the formula?
- b What does $(1-p)$ represent in the formula?

4 For a binomial probability distribution with n , trials and probability of success p , Ned performed the following calculation:

$$\binom{n}{2} p^2 \times 0.7^4$$

- a State the probability of success.
- b State the number of trials.
- c What probability is Ned trying to find with this calculation?

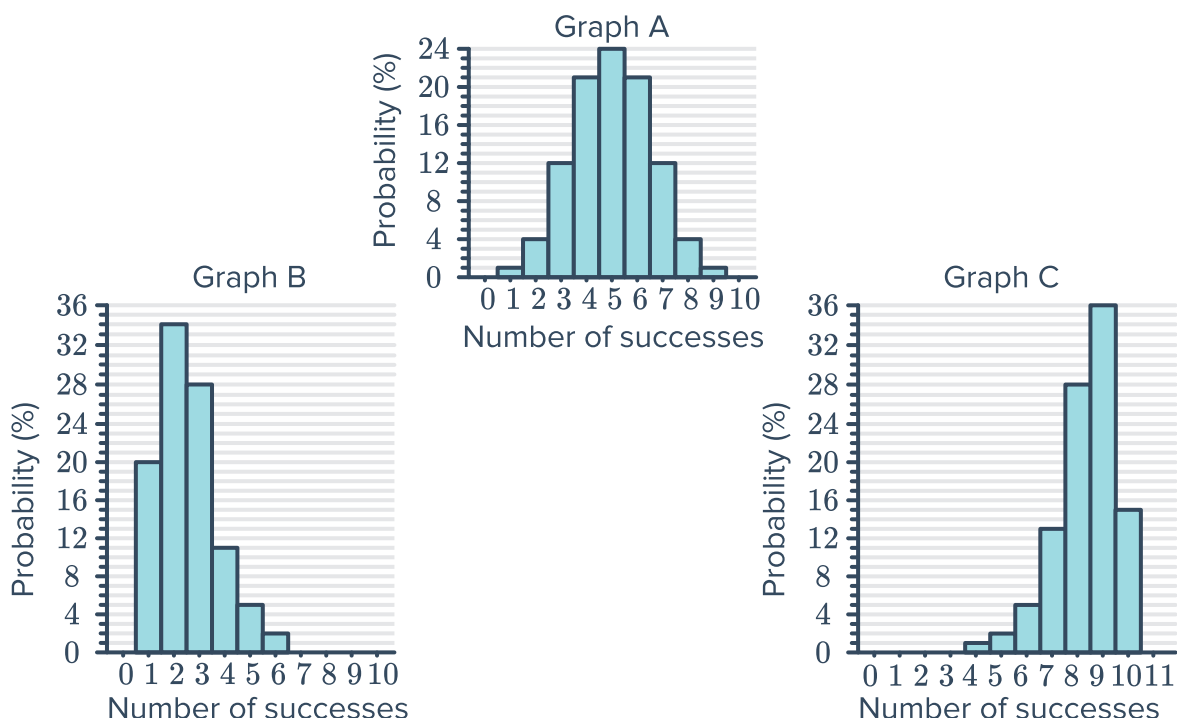
5 Consider the binomial power $(p+q)^4$.

- a Expand $(p+q)^4$.
- b We can interpret the terms in the expansion in terms of probabilities in a binomial distribution, with p the probability of success and q the probability of failure. What probability would the term $\binom{4}{2} p^2 q^2$ in the expansion represent?



Distributions of binomial random variables

- 6 Consider the three graphs below that show binomial distributions for $n = 10$ and $p = 0.15$, 0.5 and 0.85:



- a Which value of p corresponds to Graph A?
 - b Which value of p corresponds to Graph B?
 - c Which value of p corresponds to Graph C?
- 7 Consider a binomial distribution of a random variable X with $p = 0.5$ and $n = 17$.
- a Is $P(X = 5)$ greater than, less than or equal to $P(X = 12)$?
 - b Is the shape of the graph of this binomial distribution symmetric, positively skewed or negatively skewed?
- 8 Consider a binomial distribution of a random variable X with $p = 0.13$ and $n = 11$.
- a Is $P(X = 4)$ greater than, less than or equal to $P(X = 7)$?
 - b Is the shape of the graph of this binomial distribution symmetric, positively skewed or negatively skewed?

9 Consider a binomial distribution of a random variable X with $p = 0.83$ and $n = 17$.

- Is $P(X = 2)$ greater than, less than or equal to $P(X = 15)$?
- Is the shape of the graph of this binomial distribution symmetric, positively skewed or negatively skewed?

10 For each of the following table of values for the binomial distribution X :

- Describe the shape of the graph as positively skewed, symmetric, or negatively skewed.
- Calculate the value of p correct to two decimal places.

a

x	0	1	2	3	4	5	6	7	8
$P(X = x)$	0.004	0.031	0.109	0.219	0.274	0.219	0.109	0.031	0.004

b

x	0	1	2	3	4	5	6	7	8
$P(X = x)$	0.152	0.323	0.3	0.16	0.053	0.011	0.001	0	0

c

x	0	1	2	3	4	5	6	7	8
$P(X = x)$	0	0	0.001	0.009	0.046	0.147	0.294	0.336	0.168

Probability for binomial random variables

11 Find the value of the following:

- ${}^5C_4 \times (0.1)^4 \times 0.9 + {}^5C_5 \times (0.1)^5 \times (0.9)^0$
- ${}^5C_3 \times (0.8)^3 \times (0.2)^2 + {}^5C_4 \times (0.8)^4 \times 0.2 + {}^5C_5 \times (0.8)^5 \times (0.2)^0$

12 A binomial random variable X has 13 independent trials and a probability of success of 0.4.

- Using the definition for the binomial random variable X , write the formula for $P(X = 2)$.
- Calculate $P(X = 2)$ to four decimal places.

13 For a binomial random variable X , $P(X = \text{[img alt="A small graphic showing a blue and green abstract pattern." data-bbox="475 20 525 55"]}) = {}^{10}C_3 \times 0.6^3 \times 0.4^7$.

- a Calculate $P(X = 3)$ to four decimal places.
- b State the number of trials in this experiment.
- c State the probability of success in this experiment.

14 X is a binomial variable with the probability mass function:

$$P(X = k) = {}^4C_k \times (0.4)^k \times (0.6)^{4-k} \text{ for } k = 0, 1, 2, 3, 4$$

- a State the number of trials for this distribution.
- b State the probability of success.
- c Write an expression for $P(X = 2)$.
- d How many ways can we get 2 successes in the 4 trials?
- e Calculate the probability $P(X = 2)$.

15 X is a binomial variable with the probability mass function:

$$P(X = k) = {}^3C_k \times (0.3)^k \times (0.7)^{3-k} \text{ for } k = 0, 1, 2, 3$$

Complete the table:

x	0	1	2	3
$P(X = x)$	0.343	0.441		

16 Consider the random variable X , which has distribution $X \sim B(4, 0.1)$.

- a Write an expression for $P(X = 1)$.
- b Write an expression for $P(X = 2)$.
- c Complete the following table:

x	0	1	2	3	4
$P(X = x)$	0.6561				0.0001

17 A binomial random variable X is defined as:

$$P(X = x) = {}^{20}C_x \times 0.75^x \times 0.25^{20-x} \text{ for } x = 1, 2, \dots, 20$$

- a State the number of trials in this experiment.
- b State the probability of success in this experiment.
- c Calculate $P(X = 11)$ to four decimal places.

$$P(X = k) = {}^6C_k \times (0.8)^k \times (0.2)^{6-k} \text{ for } k = 0, 1, 2, 3, 4, 5, 6$$

- a State the number of trials for this distribution.
- b What is the probability of failure?
- c Calculate the probability $P(X = 3)$.
- d Calculate the probability $P(X \leq 3)$.

19 X is a binomial variable with the probability mass function:

$$P(X = k) = {}^5C_k \times (0.4)^k \times (0.6)^{5-k} \text{ for } k = 0, 1, 2, 3, 4, 5$$

- a Calculate $P(X \geq 3)$.
- b Calculate $P(X < 3)$.

20 X is a binomial variable with the probability mass function:

$$P(X = k) = {}^5C_k \times (0.2)^k \times (0.8)^{5-k} \text{ for } k = 0, 1, 2, 3, 4, 5$$

- a State the number of trials for this distribution.
- b State the probability of success.
- c Calculate the probability $P(X > 3)$.
- d What is the most likely number of successes?
- e Calculate the mean of the distribution.
- f Calculate the variance of the distribution.

21 Consider the random variable X , which has distribution $X \sim B\left(8, \frac{4}{5}\right)$.

- a State the number of trials for this distribution.
- b State the probability of success.
- c Calculate the probability $P(X > 4)$ to two decimal places.
- d What is the most likely number of successes?
- e Calculate the mean of the distribution.
- f Calculate the variance of the distribution.

22 X is a binomial variable with a probability distribution given in the table:



x	0	1	2	3	4	5
$P(X = x)$	0.00243	0.02835	0.1323	0.3087	0.36015	0.16807

- a** Find $P(X \geq 4)$. **b** Find $P(X < 3)$.

23 The following table represents a probability distribution of a binomial random variable:

x	0	1	2	3	4
$P(X = x)$	0.2401	0.4116	0.2646	0.0756	0.0081

- a** State the number of trials for this distribution.
b Calculate the probability $P(X = 3)$.
c Calculate the probability $P(X < 3)$.
d Using $P(X = 4)$, calculate the probability of success p .
e Calculate $E(X)$.
f Calculate the standard deviation of the distribution.

24 The table represents a cumulative probability distribution of a binomial random variable:

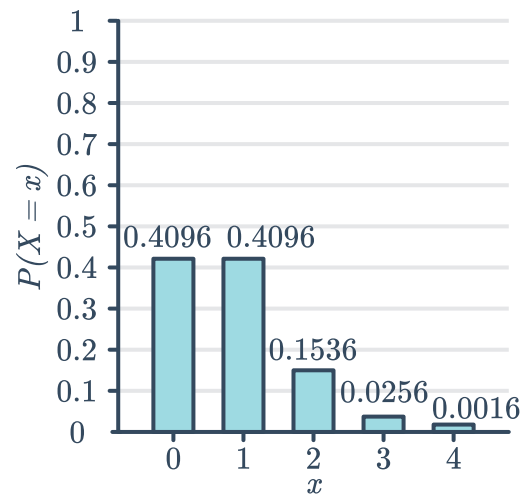
x	0	1	2	3	4	5
$P(X \leq x)$	0.01024	0.08704	0.31744	0.66304	0.92224	1

- a** State the number of trials for this distribution.
b Calculate $P(X \leq 2)$.
c Calculate $P(X \geq 3)$.
d Calculate $P(X = 5)$.
e Using $P(X = 5)$, calculate the probability of success p .
f Calculate the standard deviation of the distribution.

25 The following graph represents a probability distribution of a binomial random variable:

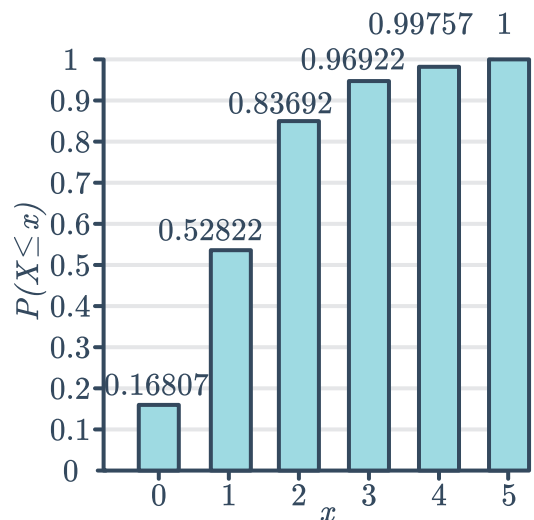


- State the number of trials for this distribution.
- Calculate $P(X = 3)$.
- Calculate $P(X < 3)$.
- Using $P(X = 4)$, calculate the probability of success p .
- Calculate $E(X)$.
- Calculate the standard deviation of the distribution.



26 The following graph represents a cumulative probability distribution of a binomial random variable:

- State the number of trials for this distribution.
- Calculate $P(X \leq 2)$.
- Calculate $P(X \geq 3)$.
- Calculate $P(X = 5)$.
- Using $P(X = 5)$, calculate the probability of success p .
- Calculate the standard deviation of the distribution.



27 Calculate the probability of the following, rounding your answers to two decimal places:

- Getting exactly 18 successes in 30 independent trials if the probability of success in each trial is 0.7.
- Getting at least 10 successes in 80 independent trials if the probability of success in each trial is $p = 0.1$.
- Getting at most 22 successes in 25 independent trials if the probability of success in each trial is 0.85.
- Getting more than 15 successes in 33 independent trials if the probability of success in each trial is 0.45.
- Getting less than 12 successes in 50 independent trials if the probability of success in each trial is 0.28.

28 $P(X = 2) = 0.3456$ for a binomial random variable X with 4 trials.



- a** If success is more likely, find the probability of success p .
- b** Calculate $E(X)$.
- c** Calculate $\text{Var}(X)$.

29 $E(X) = 6$ for a binomial random variable X with 10 trials.

- a** Find the probability of success p .
- b** Calculate $\text{Var}(X)$.

30 The expected value of a binomial distribution is 8 and the variance is 6.4.

- a** Find the number of trials, n .
- b** Find the probability of success, p .

31 $E(X) = \frac{15}{2}$ for a binomial random variable X with probability of success $\frac{3}{10}$.

- a** Find the number of trials n .
- b** Calculate the standard deviation of X .

32 $E(X) = 2$ and $\text{Var}(X) = 1.6$ for a binomial random variable X .

- a** Find the probability of success p .
- b** Find the number of trials n .

33 $E(X) = 4.8$ and $\sigma(X) = \sqrt{2.88}$ for a binomial random variable X .

- a** Find the probability of success p .
- b** Find the number of trials n .

Applications

34 A certain disease has a survival rate of 64%. Of the next 110 people who contract the disease, how many would you expect to survive? Round your answer to the nearest whole number.

35 The probability of a particular tennis player getting their first serve in is 0.72. If she serves 90 times in a match, how many times would she expect to get her first serve in? Round your answer rounded to the nearest whole number.

36 A fair die is rolled 5 times and the number of fours noted.

- a** On each trial, what is the probability of success?
- b** Calculate the probability of rolling exactly 2 fours, correct to four decimal places.



- 37** David owns 3 yellow mugs and 8 blue mugs. Each morning, he chooses a mug at random for his cup of tea before work, 5 days a week. The mug is washed and returned before the next day.
- a** Each day, what is the probability of choosing a yellow mug?
 - b** What is the probability that he used a yellow mug exactly twice this past work week? Round your answer to four decimal places.
 - c** What is the probability that he never used a yellow mug this past work week? Round your answer to four decimal places.
- 38** At a local Italian restaurant, 16 people are in line to order. The probability that any one person will choose pizza is 23%.
- a** What is the probability that exactly 4 people choose pizza? Round your answer to four decimal places.
 - b** What is the probability that exactly half of the people choose pizza? Round your answer to four decimal places.
- 39** Consider the binomial $\left(\frac{1}{5} + \frac{4}{5}\right)^5$.
- a** Write the formula for the term of the expansion that contains $\left(\frac{4}{5}\right)^3$.
 - b** A box of chocolates is $\frac{1}{5}$ dark chocolate and $\frac{4}{5}$ milk chocolate. If 5 chocolates are selected at random with replacement, describe what the term from part (a) would tell us.
 - c** Calculate the probability of selecting exactly 1 dark chocolate.
 - d** Calculate the probability of selecting at most 1 dark chocolate.